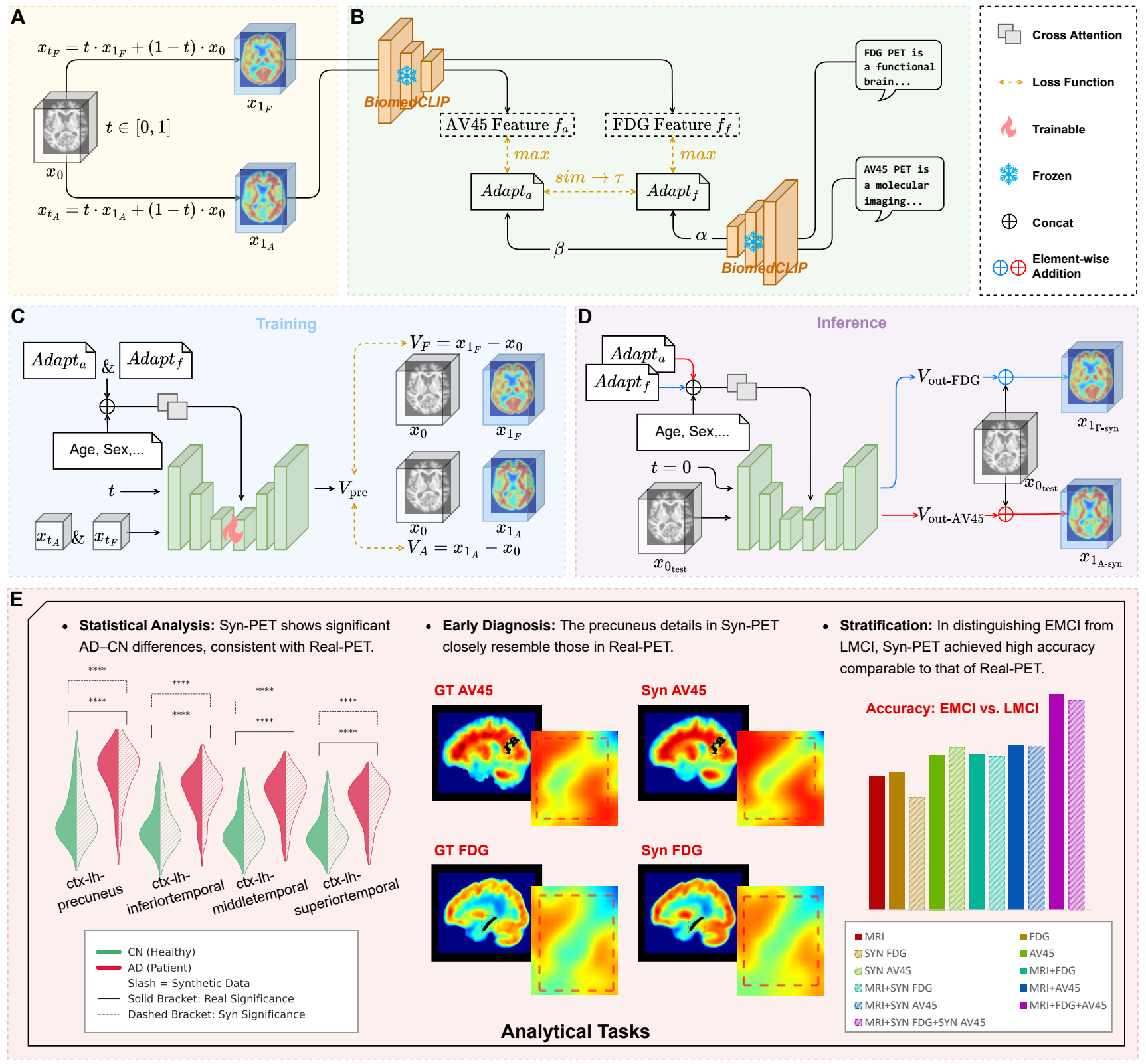

A
$x_{t_F} = t \cdot x_{1_F} + (1-t) \cdot x_0$
x_{1_F}
$t \in [0,1]$
x_0
$x_{t_A} = t \cdot x_{1_A} + (1-t) \cdot x_0$
x_{1_A}
B
BiomedCLIP
AV45 Feature f_a
FDG Feature f_f
max
max
sim → τ
$Adapt_a$
$Adapt_f$
α
β
BiomedCLIP
FDG PET is a functional brain...
AV45 PET is a molecular imaging...
Cross Attention
Loss Function
Trainable
Frozen
Concat
Element-wise Addition
C
Training
$Adapt_a$ & $Adapt_f$
Age, Sex,...
t
x_{t_A} & x_{t_F}
V_{pre}
$V_F = x_{1_F} - x_0$
$V_A = x_{1_A} - x_0$
x_0
x_{1_F}
x_0
x_{1_A}
D
Inference
$Adapt_a$
$Adapt_f$
Age, Sex,...
t = 0
$x_{0_{\text{test}}}$
$V_{\text{out-FDG}}$
$V_{\text{out-AV45}}$
$x_{1_{\text{F-syn}}}$
$x_{0_{\text{test}}}$
$x_{1_{\text{A-syn}}}$
E
Statistical Analysis: Syn-PET shows significant AD–CN differences, consistent with Real-PET.
Early Diagnosis: The precuneus details in Syn-PET closely resemble those in Real-PET.
Stratification: In distinguishing EMCI from LMCI, Syn-PET achieved high accuracy comparable to that of Real-PET.
****
ctx-lh-precuneus
ctx-lh-inferiortemporal
ctx-lh-middletemporal
ctx-lh-superiortemporal
CN (Healthy)
AD (Patient)
Slash = Synthetic Data
Solid Bracket: Real Significance
Dashed Bracket: Syn Significance
GT AV45
Syn AV45
GT FDG
Syn FDG
Accuracy: EMCI vs. LMCI
MRI
FDG
SYN FDG
AV45
SYN AV45
MRI+FDG
MRI+SYN FDG
MRI+AV45
MRI+SYN AV45
MRI+FDG+AV45
MRI+SYN FDG+SYN AV45
Analytical Tasks